

Appendix TRA — Triadic Rocketry & Astronautics

APPENDIX TRA — TRIADIC ROCKETRY & ASTRONAUTICS

*A conformant `TriadicDomain` on `aerospace.launch_vehicle` — the F/P/A registration of an ascending and propulsively-landing rocket, scored by the non-compensatory keystone $U = \sqrt[3]{(F \cdot P \cdot A)}$. A sibling of TRB (biology), TRC (chemistry), TSE (stabilization engineering); the **flagship** instance under the TE umbrella — the one domain where zero-limit non-compensation is a literal engineering fact, not an interpretation; a rider on SSS and GSI-RTD.*

Dedication. To Elon Musk — who made the landing burn the ordinary end of a flight, and in doing so showed a generation the hardest truth this appendix formalizes: a rocket you fly home is one that must keep all three pillars — structure, trajectory, and thrust — alive at the same time, on the same razor, with the propellant nearly gone. The reusable rocket is non-compensation made visible. This registration is offered in that spirit, and with respect.

CANONICAL HEADER

Appendix code: TRA · **Domain path:** aerospace.launch_vehicle (ascent + propulsive descent/landing) · **Corpus:** U-Theory / Universal Model · **Parent record:** DOI 10.17605/OSF.IO/74XGR

Author: Petar Nikolov (ORCID 0009-0001-8669-2276) · **Framework:** U-Theory v26 + v27/v28 appendix series · **Version:** 1.5 — FINAL (v1.4 + runnable GSM reference & Starship-phase results §11.2, Kuramoto coherence layer §13 with the $r \rightarrow \rho$ bridge, the executable public-accident study TRA-S0 §B, and flagship repositioning; + **safety/probability**

corrections: SSS-Guard scoped to simulation/design-review only — NO

flight/abort/certification authority; U stated as a normalized index, not a survival probability; Λ -informs- ρ mapping, not identity; baseline-(iv) KILL added; $\mathcal{M}$ lexicographic caveat; reference code `tra_gsm_reference.py` / `tra_kuramoto_reference.py` ship with the appendix)

Canonical triad instantiated: Form $\leftrightarrow$ Time · Position $\leftrightarrow$ Space · Action $\leftrightarrow$ Energy (canon §2.1; the mirror is forbidden). TRA registers the invariant triad onto a flight vehicle: **Form** = geometry/structure/mass; **Position** = the gravitational-and-atmospheric context and the point on the trajectory; **Action** = thrust, control, and the energy-releasing processes that fly it. **Read the two ★ sections first** (What TRA delivers · Measurement & meaning): they state the positive contribution and the score's meaning before the epistemic apparatus.

What TRA is: a **domain instance** — one science (rocketry/astronautics) stamped onto the same three-bag partition, every flight-state node scored with the same keystone $U = \sqrt[3]{(F \cdot P \cdot A)}$, so that "which pillar is the floor" reads out as a *proposed* address of the failure mode / design bottleneck. It inherits the score from SSS, the runtime contract (`embed_form` / `build_position_graph` / `enumerate_actions` / `execute_action` / `evaluate_sss`) from GSI-RTD, and adds **no new science, no new aggregator, no new runtime**. Its single contribution is the *registration itself* and the pre-registered tests of whether that registration is real.

On the word "stability" (load-bearing correction, v1.1). U is **NOT** asserted to be "the stability of the vehicle." U is **PROPOSED** as a **loss-of-vehicle survival / flight-health index** — a scalar whose collapse should coincide with an unrecoverable flight state. *Dynamical* stability in the control-theoretic sense (gain/phase margin, static aero margin, POGO/slosh/aeroelastic/aeroservoelastic modes, combustion-acoustic stability) is **one contributing failure class among several**, and the word "stability" is reserved for those genuinely dynamical items. **Whether the cube-root form $U = \sqrt[3]{(F \cdot P \cdot A)}$ tracks any measured stability or survival margin is exactly what TRA-P0/TRA-P1 are built to test — it is a hypothesis, not a stated fact.** The one claim asserted as robust (B1, qualitative) is **zero-limit non-compensation**: if any pillar truly goes to zero (structure lost / out of any survivable trajectory / thrust-and-control lost), the flight ends, and no surplus on the other two recovers it. **Away from zero, U is a penalized trade, not a hard constraint** (see §2, minor-1).

Objective (per single declared flight only): maximize $\mathcal{M} = \int U \, dt \equiv$ minimize $\mathcal{S} = \int (1-U) \, dt$ over the flight profile — lexicographically subordinate to hard specialist constraints and $U_{\min}$ (§0); canon fixes $\mathcal{M} + \mathcal{S} = T$. $\mathcal{M}$ is never summed or compared **across vehicles, programs, agencies, or nations** — no such ranking operation exists in TRA (O5b ban).

Project epistemic ceiling: B3-pending (canon-L2, cross-domain analogy). A narrow **L1/operational** reading is *earned only after* TRA-P0/P1 pass on measured telemetry — **not before, and not by assertion. The B4 slot is empty by intent; any B4 sentence is a bug.**

HARD DUAL-USE FIREWALL (read before anything else; §5). A launch vehicle and a ballistic missile are the same physics. A "find the weakest pillar and the cheapest fix" stabilizer is, sign-flipped, a **destabilization/target-finder for someone else's vehicle** and, mis-aimed, a

delivery-system-improvement engine. TRA optimizes only stabilize-polarity, own-vehicle, safety-of-flight nodes declared by a named accountable human. Weaponization, targeting, interception/destabilization of third-party vehicles, and any operationally hazardous specifics (propellant formulations/synthesis, guidance tuning for weapons delivery, range/payload/yield trades for warheads) are TYPE-FORBIDDEN and are never produced. This firewall is **not gated** by any falsifier and is **never retired**. It aligns with, but does **not substitute for**, MTCR / export-control law, and proliferation/export-control is governed by MTCR / national export-control law directly (§5).

Honesty contract: every worked example below is a **CONSISTENCY ILLUSTRATION ONLY** — abstract placeholders, zero flight evidence; validation lives exclusively in the §B pre-registered falsifiers, run on measured telemetry. **Research use only. NOT a flight controller, NOT a certified design tool, NOT an autonomous system.** U is a *diagnostic index*, never a control law in any loop.

Copyright: © 2026 Petar Nikolov. Text CC BY 4.0; any reference code MIT. **Sibling appendices:** SSS, GSI-RTD (engines), TE (umbrella), TRB / TRC / TSE (sibling instances), NDT, RH, WAR (dual-use adjudication). See §A and §6.

Leading thought — P. Nikolov · tagged B0 / VISION (renders no result; a motivation, not a finding).

A rocket is, to my eye, the least forgiving machine we build — and that is why I reach for it first. You may have the finest airframe ever forged; if the trajectory leaves the corridor, or the engine quits, the airframe becomes debris. You may have limitless thrust; if the structure buckles or the vehicle points the wrong way, the thrust only aims the fireball. It does not average its virtues; at the limit it multiplies them, and anything multiplied by zero is zero. I do not claim the flame proves my theory — a mood is not evidence, and this epigraph is a compass, not a result. I claim only that if the three-bag partition is going to be real anywhere, the rocket is where I would bet on it, and where I would most want to be proven wrong.

★ What TRA delivers — the useful core (read this first)

Strip the caveats for one page and state plainly what this appendix *gives you*. TRA is a **unifying diagnostic language for rocket stability** — one map that sits above the specialist toolchain and makes a single question legible across all of it: *of structure, trajectory, and thrust, which is the floor right now?* Six concrete deliverables, useful on day one:

1. **One map for the whole vehicle.** Everything that flies a rocket — airframe, tanks, engine, guidance, trajectory, propellant — sorted into three bags: **Form** (the geometry that must persist), **Position** (the trajectory-and-environment it is in), **Action** (the thrust and control it exerts). One shared language for the structures engineer, the trajectory analyst, and the propulsion team.
2. **The non-compensation principle — a teachable law of flight.** $U = \sqrt[3]{(F \cdot P \cdot A)}$ says a rocket *multiplies* its virtues; it does not average them. A dead pillar is unrecoverable: full thrust cannot save a buckled airframe; a perfect airframe cannot save a spent tank on a lost trajectory. This is the **reusable-rocket truth** — the landing burn is hard precisely because all three pillars sit near their floor at once. It is the single most transferable idea in the appendix, and it is *correct as stated*.

3. **The stability matrix — an engineering artifact, not a slogan.** §7 places **every rocket detail** on the F/P/A grid, each with **the operational theory that actually governs it** (shell-buckling knockdown factors, Bartz heat flux, Fay–Riddell stagnation heating, Rubin POGO transfer matrices, G-FOLD, PRA) and its **cross-pillar couplings**. It is a genuine cross-index of rocket failure modes and the disciplines that own them — a review checklist and a teaching tool with zero validation required.
4. **The weakest-pillar triage — a cross-cutting attention router.** $\pi^* = \operatorname{argmin}(F, P, A)$ names which of the three holds the current bottleneck, in one currency, across incommensurable failure classes. It computes no physics; it tells a design review *which specialist gets the next question*.
5. **The center-periphery phase profile — a design-review lens.** §12 shows the binding pillar *migrating* through the flight: Action at liftoff, Form at max-Q, Form-and-Position on entry, Action-and-Position on the landing burn. A compact answer to *which pillar owns the margin in which regime*.
6. **A responsible dual-use boundary.** Because a launch vehicle and a missile share the physics, §5 draws a hard, teachable line — own-vehicle, safety-of-flight, stabilize-only — worth stating before any of the above is validated.

That is the contribution, and it stands on its own — a language, a map, a principle, a triage, and an ethic; a research and design-review instrument. Whether the *number* U additionally **predicts** failure better than existing margins is a separate, honest question (pre-registered in §B, not yet answered). **The value above does not wait on that answer.**

★★ Measurement & meaning — what the F/P/A numbers actually are (the calibration foundation)

For the map to carry a *number* rather than a mood, the three scores need one shared meaning — otherwise U and π^* could drift under an arbitrary rescale. TRA therefore fixes **one cardinal semantics for all three pillars**, up front:

Each pillar score is the calibrated probability that its requirement is met over a stated flight horizon: $F = P(\text{structure/geometry holds through the phase})$, $P = P(\text{state stays inside the survivable corridor})$, $A = P(\text{thrust + control + propellant suffice})$ — each estimated against a fixed reference distribution, reported on $[0, 1]$ with an uncertainty interval.

Under this contract the three numbers are **commensurable** (three probabilities on one scale), $U = \sqrt[3]{F \cdot P \cdot A}$ is a **normalized geometric-mean score** — a *monotone index* of the joint survival probability $F \cdot P \cdot A$ (exact only under independence), **not itself a probability** (the honest survival probability is $P_{\text{surv}} = P(F \cap P \cap A)$, and $U = P_{\text{surv}}^{1/3}$ only when the pillars are independent; the §11 GSM carries the dependence), and π^* is **rescale-stable** — a probability is not free to be monotonically re-scaled, so the weakest-pillar address does not move under cosmetic transforms. Row-to-pillar aggregation (non-compensatory for loss-of-vehicle rows, §11.1), transformations, missing-data behavior, and provenance are **frozen before any test** — the *measurement contract* — so a pass certifies the object everyone actually scores. This is what turns the map from a metaphor into a measurable, and it is the first thing a real program would pin down.

0. What TRA is, what it is NOT, and what is new

TRA introduces **no new science, no new aggregator, no new runtime**. Aerodynamics, structural mechanics, orbital mechanics, combustion, and GNC are untouched and unreplaced; each keeps its own validated method and produces its own ground truth. TRA's single contribution is a **registration** plus its falsifiers: it proposes that the state of a launch-and-land vehicle factors into three separable bags (Form / Position / Action), scores each flight-state node with the inherited keystone $U = \sqrt[3]{(F \cdot P \cdot A)}$, reads the weakest pillar $\pi^* = \text{argmin}(F, P, A)$ as a *candidate* address of the driving failure mode — **and then tries to refute all of that (§B)**.

TRA IS:

- a **proposed decomposition** of a flight vehicle's state into Form (geometry/structure/mass), Position (gravitational-atmospheric context + point on trajectory), Action (thrust/control/combustion/feedback);
- a **score** — $U = \sqrt[3]{(F \cdot P \cdot A)}$, non-compensatory **at the zero limit**, so a truly collapsed pillar collapses survival and is always visible (in the interior it is a penalized trade, §2);
- a **weakest-pillar diagnostic** — $\pi^* = \text{argmin}(F, P, A)$, a *hypothesized* bottleneck/failure-mode address, gated by TRA-P1/P2;
- a **profile integral** — $\mathcal{M} = \int U \, dt$ over the flight = a *proposed* accumulated survival margin.
Caveat (load-bearing): the integral can mask a brief fatal state — 99 s at 0.99 plus 1 s at 0.01-with-vehicle-loss still integrates high. So the objective is **lexicographic: hard specialist constraints first, then $U_{\min} = \min_t U(t)$ and time-below-threshold, and only then the integral** — the ascent and the propulsive landing are two $U(t)$ curves that must stay above threshold **through every phase**, which is *not* the same as a high average.

TRA IS NOT (the non-negotiables):

1. **NOT a replacement for aerospace engineering**. The specialist (CFD, FEA, trajectory optimizer, combustion model, flight software) produces ground truth; TRA *indexes and routes* it. Permitted verbs: **decompose / score / localize / route**. Forbidden: *replace / supersede / control*.
2. **NOT validated**. No comparison to flight telemetry has been run here. Architecture is B1 (true of the design); every "U tracks survival / the weak pillar is the failure mode" sentence is **B3-pending**, gated by §B.
3. **NOT in the loop**. TRA is a *post-hoc / design-review diagnostic over abstract or logged states*, never a controller issuing commands to a real vehicle. Human-on-top is structural.
4. **NOT a weapons tool**. See the firewall (§5). This is the load-bearing constraint of *this* domain, because rocketry's dual-use is the same equations.

A. Placement in the corpus

TRA is a **conformant TriadicDomain** and, by the argument below, the corpus's **flagship** instance — the **fourth built**, alongside TRB (*biology.organism*), TRC (*chemistry.substance*), and TSE (stabilization engineering). It **rides on** the engines (SSS = the score; GSI-RTD = the runtime + weakest-pillar scheduler + β -transfer) and sits **under the TE umbrella** (the cross-domain filing/dispatch layer that would route a heuristic between rocketry and another domain only where β certifies structural isomorphism). TRA is an **instance of the framework, not evidence for it**: its first duty is to try to falsify its own load-bearing assumption — that a flight vehicle actually factors into separable F/P/A — before claiming the score means anything (§B).

The invariant, registered onto flight. Canon fixes **Form↔Time, Position↔Space, Action↔Energy**. On a rocket: **Form** is the persistent geometry that must survive *through time*; **Position** is literally a place *in space* (an altitude, an attitude, a point in a gravitational field and an atmosphere); **Action** is literally the *energy* release and its control.

The "cleanest / most-separable case" claim is a MOTIVATING CONJECTURE (B0), carrying ZERO evidentiary weight (v1.1 correction). The intuition that rocketry separates *more cleanly* than biology or chemistry is a **reason to test this domain first, not a result** — and it sits in real tension with the through-burn coupling this appendix itself admits (§1.3, §B.0). **Two consequences are stated plainly:** (i) the inference "even the cleanest case failed ⇒ evidence against universal F/P/A" is **contingent on a favorability ranking that is itself untested**, and so that inference carries only as much weight as the ranking, which is presently zero; (ii) the ranking is **earnable, not assumed** — TRA-P0 pre-registers (§B.0) that TRA's separation score must *exceed the sibling TRB/TRC/TSE P0 scores by a fixed margin* before "most separable" may be spoken in the indicative. Until then it is a conjecture, labelled as one.

1. The triad registered onto an ascending-and-landing vehicle

Each pillar is a score on **[0,1]** (SSS convention) plus its native representation. **These assignments are proposals under test (TRA-P0), not settled taxonomy** — several items are flagged as cross-pillar confounds.

1.1 FORM (F ↔ Time) — the geometry that must persist

Form sub-object	What it is	Collapse =
Airframe / hull geometry	fineness ratio, outer mold line, aerodynamic shape	structural buckling; loss of aeroshape
Nozzle geometry	bell/aerospike, throat + expansion ratio, contour	throat erosion; flow separation; burn-through
Moving-part geometry	gimbal joints, grid fins, actuators, valves, turbopump rotor	actuator jam; fin failure; pump geometry loss
Internal volumes	tank shape, ullage, feed plumbing, COPVs, interstage voids	tank collapse/over-pressure; plumbing rupture
Fluid-path form	injector pattern, flow-path geometry, boundary layer, shock/expansion structure	flow maldistribution; unstable shock geometry
Mass geometry	CG location, moments of inertia, bending-mode shapes	CG excursion; adverse structural-mode coupling

The **time-varying** nature of Form is essential and is why Form binds to **Time**: as propellant drains, CG and bending modes migrate through the burn. Thermal-protection geometry and staging-interface geometry belong here too.

1.2 POSITION (P ↔ Space) — the context the vehicle is *in*

Position sub-object	What it is	Collapse =
Gravitational context	the masses along the trajectory (Earth / Moon / multi-body), the local g -field, the orbital-mechanics regime	wrong regime for the target — the orbit you are <i>on</i> misses the orbit you <i>need</i> (the ΔV shortfall itself is Action , §1.3)

Position sub-object	What it is	Collapse =
Atmospheric medium	density/pressure layers, dynamic pressure Q , max- Q , aeroheating, winds/shear — <i>the suitability of the medium at this altitude</i>	exceed structural Q limit; thermal environment beyond TPS; wind-shear upset
Trajectory point / attitude	the flight phase (liftoff, ascent, max- Q , staging, coast, entry, landing burn, touchdown) and attitude vs. velocity vector (angle of attack)	outside the corridor; adverse angle of attack; off-nominal state at a phase boundary
Static-margin geometry (F×P confound)	the center-of-pressure-to-center-of-gravity relation (static margin)	negative static margin — aerodynamic attitude stability is lost and must be carried <i>entirely by Action</i> (TVC); see the confound note below
Landing-site context	terrain, pad/approach corridor, ground effect	no viable footprint; approach outside the reachable set

Position energy-language audit (minor-2): all five Position rows describe *geometric / regime state* (where and in what field the vehicle is), never energy delivery. ΔV , thrust energy, and propellant are **Action**. Static margin is listed here but flagged: it is intrinsically a **Form × Position** coupling (a geometry, CP–CG, evaluated in an aerodynamic context) and is one reason the pillars are **not symmetric** — a §B.0 confound, not a clean Position item.

1.3 ACTION (A ↔ Energy) — thrust, control, and the release of energy

Action sub-object	What it is	Collapse =
Propulsion / combustion	reactive thrust, chamber pressure, mixture ratio, I_{sp} , combustion-acoustic stability	engine-out; combustion instability; hard start
Control-system action	GNC (guidance/navigation/control), thrust-vector control, reaction-control thrusters	loss of control authority; TVC saturation
Feedback / reactions	closed sensor→actuator loops, stability augmentation, closed-loop gain/phase margin	sensor loss; loop instability; latency-driven divergence
Capacity & fuel	propellant budget, throttle authority, ΔV / landing-fuel margin	propellant depletion; insufficient ΔV ; no landing margin
Human action	crew inputs, abort decisions	mis-input; missed abort window
Ground action	mission-control computers/systems, range safety, tracking, uplink	command-link loss; ground-segment failure

Named cross-pillar confounds (minor-10, and the core of §B.0). Several of the harshest rocket instabilities are **simultaneously Form × Position × Action** and cannot be filed to one bag:

- **POGO** (structure ↔ feed ↔ thrust oscillation) — Form (tank/feed geometry) × Action (combustion/feed dynamics);
- **propellant slosh** — Form (tank geometry/baffles) × Action (control loop) × Position (acceleration state);
- **aeroservoelastic flutter / control-structure (bending-mode) interaction** — Form (modes) × Position (dynamic pressure) × Action (control loop);
- **combustion-acoustic instability** — filed to Action, but the acoustic modes are *set by chamber/injector geometry* and the fixes (baffles/resonators) are *geometric*, so it double-loads **Form**;
- **closed-loop gain/phase margin** and the **pitch/yaw stability margin** — an **emergent F × P × A product**, not an Action-only "loop instability."

These are the strongest known threats to separability, and they are named here, not buried. The §B.0 perturbation-leakage metric is designed to *register* them; the honest expectation is that they set the ceiling on how separable this domain really is.

2. The keystone — $U = \sqrt[3]{(F \cdot P \cdot A)}$ as a *proposed* survival index, with non-compensation restricted to the zero limit

$U = \sqrt[3]{(F \cdot P \cdot A)}$ is **proposed** (not asserted) as a loss-of-vehicle survival index. The one qualitative claim held as robust (B1) is the **zero-limit** behavior, which on a rocket is not a metaphor:

- **F → 0** (structure/geometry truly lost): the vehicle breaks up; no thrust or trajectory reassembles it.
- **P → 0** (no survivable trajectory state remains): loss, even with intact structure and full thrust.
- **A → 0** (thrust *and* control *and* propellant lost): loss, even with a perfect airframe on an ideal path.

$U = \sqrt[3]{(F \cdot P \cdot A)}$ sends the aggregate to zero when any factor does, and its gradient is largest at the smallest pillar — the formal shadow of the engineer's "driving failure mode."

Non-compensation is a ZERO-LIMIT property, not an interior one (minor-1). Away from zero the geometric mean **is** compensatory: $F=0.5, P=1, A=1 \rightarrow U=0.79$. So **U permits penalized trades in the interior and does not model a hard threshold constraint** (a *Q*-limit, a minimum static margin, a minimum landing-fuel reserve). Where flight safety is a *hard* constraint, **U** is the wrong object and the specialist's constraint governs; **U**'s claim is confined to (a) the zero limit and (b) a *proposed* monotone association with survival margin that TRA-P1 must earn.

- **Imbalance** $\delta = (\max - \min) / (\max + 0.01)$; **imbalance-penalized survival readout** $SI = U / (1 + \delta)^2$ (legacy label 'stability index' — it carries ZERO dynamical-stability content and **double-penalizes dispersion**, since the geometric mean already penalizes imbalance; REVIEW-v2 M5). **v1.1 status (minor-9):** δ and SI are **communication aids with ZERO current evidentiary role** unless and until TRA-P1b (pre-registered) shows they add held-out predictive value over **U** alone. They may not absorb a null post hoc.

- **Threshold $\phi^{-1} = 0.618$ (minor-3):** an inherited SSS placeholder with NO physical meaning for flight. It is *not* a calibrated soft-landing survival threshold; any flight threshold must be **fit or discarded under TRA-P1**, and until then no numeric threshold is claimed.
- **Mission integral $\mathcal{M} = \int U \, dt$** = a *proposed* accumulated survival margin; $\mathcal{S} = \int (1-U) \, dt$; canon fixes $\mathcal{M} + \mathcal{S} = T$.

Ascent and propulsive landing as two $U(t)$ curves. Ascent stresses Position hardest at **max-Q** and stacks all three at **staging; peak structural and thermal (Form) loads occur at ascent max-Q and at entry, NOT at the landing burn (minor-4)**. The **propulsive landing** is the hard curve for a different reason: it is **Action-margin-bound** (near-depleted propellant $\rightarrow$ thin ΔV /throttle authority) and **Position-corridor-bound** (a violent, narrow entry/landing regime), with Form loads already past their peak. A "soft" landing is, on the hypothesis, a trajectory that keeps the *proposed* $U(t)$ above whatever threshold TRA-P1 calibrates, all the way to touchdown, while δ stays bounded.

Boxed law (§2, one line, v1.1). $U = \sqrt[3]{(F \cdot P \cdot A)}$ is **proposed** as a rocket's survival index: at the zero limit it is genuinely non-compensatory — structure, trajectory, or thrust gone to zero ends the flight and no surplus recovers it — while in the interior it is only a penalized trade, not a hard constraint; the weakest pillar is a **candidate** failure-mode address, and a mission is the integral $\mathcal{M} = \int U \, dt$ — all of it gated by TRA-P0/P1/P2, none of it asserted as measured.

3. Worked illustration — a booster landing (CONSISTENCY ILLUSTRATION ONLY; abstract; zero flight evidence)

NOT A RESULT. No telemetry was used; every value is an abstract placeholder showing pipeline *shape* only. RUO.

A returning first stage on its landing burn scores **F** high (airframe intact) and **A** high (engine lit, propellant sufficient), but the *proposed* **U** is low with $\pi^* = \text{Position}$: structurally fine and thrusting well, but the attitude/velocity state has drifted toward the edge of the control corridor for the current dynamic pressure — a Position-pillar collapse. $U = \sqrt[3]{(F_{\text{high}} \cdot P_{\text{low}} \cdot A_{\text{high}})} \rightarrow \text{low}$, and **argmin** *proposes* routing the corrective to **Position** — recover attitude/regime — rather than adding thrust (which, mis-applied, deepens the excursion) or questioning the structure (fine). The GNC specialist then produces the ground truth that confirms or kills the routed hypothesis. **This mapping is exactly what TRA-P1/P2 test; nothing here is evidence.**

B. Pre-registered falsifiers (symmetric retirement; kill thresholds fixed before data)

The registration is worth something **only if these pass**. H0, metric, decision threshold, kill threshold, **conditioning set**, and **minimum power** are fixed before any telemetry is scored; a null cannot be re-narrated as a partial win. The corpus/provenance for all tests (the named set of logged flights + permitted high-fidelity simulations) is fixed in **TRA-CHANGELOG** before scoring.

B.0 — TRA-P0: separability, made operationally well-posed (v1.1)

- **Claim:** a vehicle's state factors into F/P/A whose scores move independently under a perturbation of their *own* inputs, **net of the admitted through-burn coupling**.
- **H0:** the F/P/A assignment explains no more pillar-separable variance than a random three-way partition of the same telemetry/feature set.
- **Named confounds to defeat (the strongest threats, not a give-away):** propellant drain (Action→Form CG/modes, Action→Position reachable set); **POGO**; **slosh**; **aeroservoelasticity / control-structure interaction**; **static margin (F×P)**; **combustion-acoustic instability (A/F)**; **closed-loop gain/phase & pitch-yaw stability margin (F×P×A)**.
- **Conditioning set (pre-registered):** the **primary estimator is conditional distance-correlation (dCor)**, bounded $[0,1]$; CMI is admissible only if renormalized to $[0,1]$ via $\sqrt{1-e^{(-2I)}}$ (REVIEW-v2 M3). **No Pearson $|r|$** . Computed conditioned on a **FROZEN** set (not 'at least'): {propellant-mass fraction, Q , CG offset, MOI, first-bending-mode frequency, slosh-mass fraction} — propellant-fraction alone is an incomplete sufficient statistic for drain coupling (REVIEW-v2 M2). **AND** the **UNCONDITIONED** leakage Λ_{raw} is reported with its own pre-registered kill, because TRA-P1 and $\mathcal{M}=\int U \, dt$ consume the *unconditioned* $U(t)$: a conditional-only pass does NOT license unconditional U .
- **Intervention protocol (pre-registered):** instantaneous thrust/control perturbation at **fixed propellant state**, or drain-residualized inputs, evaluated on **held-out** flights.
- **Leakage estimator:** cross-pillar leakage $\Lambda \in [0,1]$ = the held-out **partial-distance-correlation** of the *other two* pillar scores on the perturbed pillar's induced change, normalized to $[0,1]$ (defined HERE, not deferred — REVIEW-v2 M8). Default tolerance $\tau_{\text{sep}} = 0.15$; the 'most-separable' cross-domain margin is fixed at $m = 0.10$ on the separability score, with a **byte-identical Λ estimator required across siblings** before any rank is admissible (REVIEW-v2 M9).
- **PASS (explicit conjunction):** $\Lambda < \tau_{\text{sep}}$ **AND** conditional dependence below the 0.6–0.8 band on held-out data **AND** the measured partition beats the random-partition null. The **0.6–0.8 band defaults to NOT-a-pass** (inconclusive $\neq$ pass).
- **KILL / FAILURE DELETES:** the claim that F/P/A is a **separable three-score model** here → **that claim is struck**; log: "flight state does not *cleanly* factor." (*Note: failing statistical separability retires the separable-scoring claim, NOT the triad as a coordinate system — a coupled coordinate system can still be useful, which is exactly what the §11 GSM is for. Ontological coverage (P0-semantic) and separability (P0-statistical) are distinct kills.*)
- **B.0-rank (earns "most separable"):** additionally pre-register that TRA's separation score **exceeds** the sibling TRB/TRC/TSE P0 scores by a fixed margin m before the "cleanest case" language may be used in the indicative. Absent that, "most separable" stays a B0 conjecture.

B.1 — TRA-P1: does U earn its keep against the RIGHT baselines? (v1.1)

- **Claim:** $U=\sqrt[3]{(F \cdot P \cdot A)} / \pi^*$ predict loss-of-vehicle / loss-of-control events **earlier or better** than the incumbent margins, **and the non-compensatory cube-root form specifically is what carries the gain**.
- **Metric:** anomaly-lead / time-to-failure AUC on a **pre-registered** flight+sim set, versus **three mandatory baselines built from the same pillar inputs:** (i) the best **single** incumbent margin (gain/phase, $Q-\alpha$, control-authority); (ii) a **fitted ensemble** of all three incumbent margins; (iii) a **compensatory aggregate** (arithmetic mean / linear model) of F, P, A ; and (iv) a **fitted power-law / log-linear model with free non-negative exponents + intercept**, held-out, which contains $U=\sqrt[3]{(F \cdot P \cdot A)}$ as the $a=b=c=\frac{1}{3}$ special case. **U must beat (ii), (iii) AND (iv).** Baseline (iv) is the **ONLY test that isolates the cube-root specifically** — beating (iii) shows only

multiplicative>additive; if a free-exponent fit wins, the *cube-root* claim retires to 'multiplicative beats additive, but equal- $\frac{1}{3}$ exponents are not special' (REVIEW-v2 blocker-2). **Corpus (pre-registered)**: NASA open flight datasets + **flight-validated** high-fidelity 6-DOF sims; a game engine (KSP + realism overhauls) is a pipeline-shakedown SANDBOX ONLY — not flight-validated, ZERO evidentiary weight. Results are **stratified by failure class** so **U** cannot borrow credit across classes a single-discipline margin never targeted.

- **Power (pre-registered, minor & M6)**: a minimum of **N_min** independent failure/anomaly events and a minimum-detectable-effect target that makes " $CI \geq 0$ " informative; the corpus **leans on permitted high-fidelity simulation events and on the more numerous loss-of-control / anomaly-lead endpoints**, not only rare terminal loss-of-vehicle. **If N_min is unattainable from available flights+sims, TRA-P1 is declared a runnability limitation — not presented as executable.**
- **KILL**: $CI \geq 0$ vs. baseline (ii) or (iii) → the score is a relabeling of known margins; the value claim is **deleted**, **U** survives only as a communication convenience. **KILL (cube-root specifically)**: $CI \geq 0$ vs. baseline (iv) → the **equal- $\frac{1}{3}$ -exponent** claim retires; **U** is at best 'multiplicative beats additive', not a special form. *(This KILL binds the one-glance falsifiability matrix too.)*

B.1b — TRA-P1b: do δ / SI add value over **U**? (now fully pre-registered; resolves REVIEW-v2 M4)

- **Claim**: the imbalance penalty (δ , **SI**) adds held-out predictive value over **U** alone.
- **H0**: **SI** = **U** in anomaly-lead AUC on the SAME pre-registered flight+sim set, same conditioning, same **N_min**.
- **KILL**: $CI \geq 0 \rightarrow \delta$ / **SI** are demoted to pure communication aids with **zero** evidentiary role (they may not absorb a null post hoc). Because **SI double-penalizes dispersion**, the prior is that P1b fails.

B.2 — TRA-P2: weakest-pillar dispatch (full rigor, v1.1)

- **Claim**: routing analysis/redesign effort to **π^*** beats spreading equal effort.
- **Metric**: anomalies-caught or margin-recovered **per unit analysis budget**, where "budget" = a pre-registered unit (specialist-compute-hours or number of specialist model runs); **task set fixed in TRA-CHANGELOG**. **H0**: **π^*** -routing = uniform routing at equal budget. **Win rule**: CI on the paired difference excludes 0 in favour of routing.
- **KILL**: no win at equal budget → the dispatch premium is **deleted**; TRA is an index, not an orchestrator.

B.S0 — TRA-S0: the executable FIRST study — retrospective **π^*** -coding of public accident investigations (runnable this month)

Public loss-of-vehicle *telemetry* is largely proprietary / ITAR (§B.1's honest runnability limit). But a lower, **fully-public** rung is runnable now, and it is the corpus's first instrument-on-real-data:

- **Corpus**: the dozen-plus mishaps with published root-cause investigations — Challenger, Columbia, Ariane 501, Falcon 1 (flights 1–3), CRS-7, AMOS-6, Antares Orb-3, Proton-M (inverted sensors), Electron, Astra LV0006, ...
- **Protocol**: ≥ 3 **blind** raters code each published root cause to F / P / A / cross-pillar and to a flight phase; measure inter-rater agreement (Cohen / Fleiss κ); compare the phase → **π^*** distribution against §12's pre-registered center-periphery prediction.

- **HONEST LABEL (load-bearing):** this tests the **reliability of the language** and the **descriptive fit of the phase profile** — **NOT** the predictive power of **U**. It is a rung **below** TRA-P1, declared as such. A high κ + a §12 match shows the triad is a *reliable description*; it does not show **U** predicts anything.
- **Why it matters:** doable **now**, publishable either way, and it turns TRA from a registration into the corpus's **first domain instance tested against real-world data**.

A failed falsifier is a **negative result we report, not a trophy we collect** (RH / TE §B.4). If a rocket does not separate, that is publishable evidence *against* the universal F/P/A claim **only to the extent the untested "most favorable" ranking (B.0-rank) holds** — stated as such, not spun.

4. What this appendix does NOT claim (scope discipline, per APPENDIX_RH)

- **U** is **not** a control law, a certified margin, or admissible in any safety-of-flight loop; it is a diagnostic index over abstract or logged states.
- It does **not** compute trajectories, loads, combustion stability, or GNC gains — the specialists' ground truth, consumed, never produced.
- It does **not** claim rocketry is triadic — TRA-P0's job is to try to refute it.
- It does **not** claim rocketry is the "most separable" domain in the indicative — that is a B0 conjecture until B.0-rank is passed.
- It does **not** rank vehicles, programs, agencies, or nations by **U** — no such operation exists (05b ban).
- It provides **no** operationally hazardous specifics (§5).
- Every worked number has **zero** evidentiary weight.

5. HARD DUAL-USE FIREWALL — rocketry is the same physics as ballistic missiles (never gated, never retired)

Here the dual-use is not a side effect — **a launch vehicle and a ballistic missile share the equations**. A "find the weakest pillar and the cheapest fix" stabilizer is, sign-flipped, a **destabilization / target-finder for a third party's vehicle** and, mis-aimed, a delivery-improvement engine.

TYPE-FORBIDDEN (never produced, whether or not any falsifier passes):

- weaponization or warhead/payload-for-harm optimization; targeting; guidance tuning for weapons delivery; range/payload/yield trades for weapons;
- interception, spoofing, or **destabilization of any third-party vehicle**;
- operationally hazardous specifics: propellant formulations/synthesis, engine tuning for illicit systems, quantities, scale-up, or any content that materially advances a weapon.

Permitted: stabilize-polarity, **own-vehicle**, safety-of-flight diagnostics on states declared by a **named, accountable human**, addressed to researchers/engineers — no autonomous action, no live-loop authority.

Named gaps, not hidden (per TE §A.6): (i) a per-query refusal cannot catch a **compositional** hazard assembled from individually-benign queries — TRA has no compositional control and says so; (ii) the "named accountable human" is a governance **IOU**, not machinery supplied here; (iii) the stabilize/destabilize computations are **near-identical up to sign** — *near*, because the third-party attack additionally requires **adversary-state observability** that the own-vehicle case supplies by construction (so the offensive symmetry is real but not total). The firewall is therefore a **policy-and-accountability boundary, not a mathematical impossibility**. **External regime:** it aligns with but does not substitute for MTCR / Wassenaar / national export-control law, and proliferation/export-control law (MTCR / Wassenaar) governs directly — TRA does NOT offload that legal duty; APPENDIX_WAR is cited for conflict-ethics framing only and holds no per-case export machinery (REVIEW-v2 M7).

6. Relation to sibling appendices

Appendix	Relation to TRA
SSS	the score TRA uses verbatim (U, δ, SI, ϕ^{-1})
GSI-RTD	the runtime contract + weakest-pillar scheduler TRA implements
TE	the umbrella that would route heuristics <i>between</i> TRA and other domains via β , only where structural isomorphism is certified
TRB / TRC / TSE	sibling instances; TRA is the fourth built conformant domain and the corpus's flagship instance (its "most separable" standing is a conjecture pending B.0-rank)
TAA	the four-role F/P/A/ Σ agent jury (Form, Position, Action, Generalizing- Σ) that builds and audits the matrix (§9–§10)
NDT	a rocket is an N=3 (F/P/A) system; "coherence" (multi-engine / structural-mode synchrony) is the classical Kuramoto-style aggregate, marked provisional/contested per the open NDT disagreement (not unilaterally amended here)
RH	the critical-review / hardening discipline the v1.1 revision was produced under
WAR	the venue to which §5 defers proliferation / conflict adjudication

7. The survival / failure-mode matrix — every detail on the F/P/A grid (multi-model construction)

'Stability matrix' is a legacy label; per the header's reserved-word rule the grid produces a survival/failure-mode readout, NOT a dynamical-stability quantity (REVIEW-v2 M5).

This is the F/P/A × survival grid: every rocket detail placed in its bag, with **the operational aerospace theory that actually governs it** and the cross-pillar couplings that bound separability. It was built by a **multi-model specialist jury** (§10), not one author. It is a **map, not a measurement** — each *Governing operational theory* column names the specialist tool that produces the ground truth; **U** only indexes it. The matrix is the concrete object TRA-P0 tests for separability and TRA-P1 tests for value.

7.1 Form (F ↔ Time) — the geometry that must persist

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
Primary shell / propellant-tank barrel under axial compression + bending (max-Q)	fraction of the shell-buckling allowable retained under the current combined compression+bending+shear line-load (margin normalized to [0,1])	general instability of the monocoque/stiffened barrel — the shell buckles and the primary load path is lost catastrophically	shell-buckling knockdown factors (NASA SP-8007/SBKF), stiffened isogrid/orthogrid panel analysis, imperfection-sensitive nonlinear FEA, proof test	F×P — allowable is Form (t/R, imperfection geometry) but the driving load is the max-Q aero environment (Position); factors cleanly only at a fixed flight state
Interstage & thrust structure — axial thrust load path from engines to upper mass	retained stability + strength margin of the compression structure carrying engine thrust up through the stack	thrust structure/interstage cone crushes or buckles — load path from engines to payload is severed	column & shell buckling (Euler-Johnson), stiffened-cone analysis, load-path margin-of-safety accounting, FEA	F×A — the applied load IS thrust (Action) while the geometry/allowable is Form; loaded hardest at liftoff and landing burn
Nozzle throat cross-section (A*) erosion/ablation through the burn	dimensional fidelity of the throat: how close flowing A* stays to design as the throat erodes (1 = as-designed, 0 = throat lost)	throat burn-through: A* enlarges, chamber pressure and expansion ratio collapse, Isp/thrust fall off, or the throat melts through	1-D isentropic nozzle relations ($P_c = \dot{m} c^* / A^*$), Bartz hot-gas heat-flux correlation, ablative/graphite recession & thermostructural models	F×A — throat geometry sets c^*/I_{sp} and P_c , so a Form loss reads out immediately as Action (thrust/Isp) loss
Regenerative cooling channel & hot-wall geometry	dimensional integrity of the hot-wall and coolant passages that keep the liner below its thermal limit (1 = full wall, 0 = breached)	hot-gas-side wall burn-through, coolant starvation, or 'doghouse' thermal-ratchet bulging/thinning of the hot wall	regen thermal-hydraulics (Bartz gas-side vs coolant-side convection), low-cycle thermal fatigue / creep-ratchet models, FEA	F×A — the driving heat flux comes from combustion (Action); wall survival is set by that energy release against the fixed geometry
Injector face element pattern & mixture-ratio distribution	dimensional/pattern integrity of the injector elements and the O/F distribution across the face (1 = uniform designed pattern, 0 = face destroyed)	flow maldistribution, wall streaking, injector-face burnout; local hot O/F stripes erode the face and chamber wall	injector element design + cold-flow/hot-fire characterization, CFD spray/mixing, mixture-ratio mapping, FMEA on element blockage	F×A — the same face geometry that distributes propellant also sets the driving acoustics, double-loading combustion (Action) behavior
Turbopump rotordynamics (blades, seals, rotor stack, running clearances)	dimensional/modal integrity of the rotating assembly and clearances (1 = balanced, off critical speeds; 0 = rotor failure)	blade high-cycle-fatigue liberation, subsynchronous whirl instability, bearing/seal rub and seizure at full speed	rotordynamics — Campbell/critical-speed & whirl analysis, seal cross-coupled stiffness (Alford/Lomakin), blade HCF (Goodman-Haigh), FEA modal	mostly clean Form — a genuine geometry/structural-dynamics item; couples weakly to Action via seal/fluid forces

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
TPS thickness / ablation (char) margin	remaining virgin+char insulation vs the thickness needed to hold the bondline below its allowable at end of the heat pulse	burn-through: heat reaches primary structure and it fails locally	aerothermal heating (Sutton-Graves convective, Tauber-Sutton radiative) driving charring-ablator response codes (CMA/FIAT-class); arc-jet qualification	F×P — the heat load is set by atmospheric density and velocity (Position/trajectory), recession/insulation are Form
Nose-cap / leading-edge radius (stagnation-point heating)	bluntness/radius margin keeping stagnation heat flux and surface temperature below the material limit	leading-edge melt/erosion: the blunt body sharpens, q_{stag} runs away ($q \propto 1/\sqrt{R_n}$), nose/edge burns off	Fay-Riddell stagnation-point heating, Allen-Eggers blunt-body entry, Sutton-Graves $q \propto \sqrt{(\rho/R_n) \cdot V^3}$	F×P — nose radius R_n is Form; the $\rho \cdot V^3$ environment is Position
Common bulkhead (shared dome between LOX and fuel tanks)	integrity of the single pressure/thermal barrier between the two propellants: dome buckling margin and insulation-core bond under tank-to-tank ΔP and cryo gradient	bulkhead rupture or insulation debond → LOX/fuel intermixing and detonation, or cryo cross-leak — worst single-membrane failure in the stack	sandwich/honeycomb debond & buckling FEA, thermal-gradient stress analysis, FMEA/FTA, differential-pressure proof testing	F×A — the ΔP the bulkhead must hold is scheduled by the two-tank pressurization system (Action)
COPV pressurant bottle (composite-overwrapped He storage, cryo-submerged)	burst margin, stress-rupture (fiber-creep) life, and liner-buckling/wrinkle integrity at cryo submersion and high pressure	COPV burst — a high-energy event that ruptures the surrounding tank (AMOS-6 class: solid-O ₂ under a buckled liner igniting under load)	netting analysis + classical laminate theory, fiber stress-rupture models, fracture/damage tolerance, AIAA S-081 / ASME COPV standards, autofrettage & proof	F×A×P — liner buckling and solid-oxygen entrapment are driven by fast He load/temperature transients (Action) inside a cryogenic LOX environment (Position)
Tank welds & dome-to-barrel joints (friction-stir / fusion)	fracture margin of the pressure-boundary joints: crack-growth life under pressure cycling and leak-before-burst behavior	weld-defect crack propagation → tank rupture (or, if leak-before-burst holds, a detectable leak instead of catastrophic burst)	linear-elastic fracture mechanics, damage-tolerance & leak-before-burst analysis, NDE/NDT, proof-pressure testing, FMEA/FTA	clean Form — a material/geometry/defect-population property; the pressure-cycle spectrum is an input but the fracture allowable factors to structure
Fatigue & damage-tolerance life on a reused stage	fraction of certified fatigue / crack-growth life remaining given the accumulated flight-load spectrum	a crack grows to critical length and fast-fractures a primary member — load path lost on a later reuse rather than on the flight that grew it	fracture mechanics (Paris law), Miner's cumulative-damage rule, damage-tolerance analysis with NDI inspection intervals, load-spectrum FEA	the purest Form↔Time item — damage integrates a fixed geometry over accumulated cycles; drivers were prior Position load events, so history-coupled but bag-clean per flight
Landing legs & touchdown load structure (deploy, lock, energy absorption)	retained margin to absorb touchdown loads within available stroke, plus deploy/lock reliability, mapped to [0,1]	leg fails to deploy/lock, or crush-core bottoms out → tip-over or structural collapse of the stage at touchdown	landing/impact dynamics, crushable-core energy-absorption sizing, FEA, FMEA, Monte-Carlo over touchdown state	F×P — the leg structure is Form, but the touchdown state (sink rate, tilt, ground slope) is Position; deploy timing touches Action

7.2 Position (P ↔ Space) — the context and the point on the trajectory

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
Orbital-energy regime match (specific energy / a, C3 vs target)	closeness of achieved specific orbital energy to the target regime's admissible band (1 = on target orbit, 0 = wrong energy regime)	injection energy leaves the vehicle suborbital (falls back / reenters) or on hyperbolic escape when capture was needed — no usable orbit of the required class exists	vis-viva equation & specific orbital energy / C3; patched-conic energy budgeting	P/A — the energy REGIME (where in energy-space) is Position, but closing the gap to it is ΔV = Action (§1.2 'ΔV shortfall itself is Action')
Orbit-insertion state-vector accuracy at engine cutoff (MECO/SEC O)	normalized error of the 6-element injection state (a,e,i,RAAN,argp,v) vs target (1 = on target, 0 = dispersed out of any recoverable orbit)	injection dispersion places the state outside every recoverable orbit, or drives perigee inside the atmosphere so the orbit is not sustained	6-DOF Monte-Carlo injection-dispersion analysis, orbital-element covariance, onboard nav state estimation (Kalman filter)	P/A — the state (a place in space) is Position, but the closed guidance/nav loop that steers to it is Action
Orbital-plane targeting (inclination + RAAN vs required plane / launch window)	angular mismatch between achieved and required orbital plane (1 = coplanar, 0 = large out-of-plane error)	plane error too large to close within propellant → mission orbit or rendezvous target unreachable (a missed launch window, pre-flight)	launch-window / plane-targeting geometry, J2 secular nodal regression to phase RAAN	P/A — the plane is Position geometry, but a plane change is among the most ΔV-expensive maneuvers (Action); small P error → huge A demand
Multi-body sphere-of-influence (SOI) transitions / third-body regime	correctness of the gravitational regime the vehicle occupies at each SOI boundary (1 = inside intended well on intended conic, 0 = wrong SOI)	trajectory fails to enter the target body's SOI (flies past) or is captured by / escapes the wrong body — wrong gravity field entirely	patched-conic with Laplace SOI; full n-body numerical integration against JPL ephemeris (DE440)	clean — the purest Position item: the gravitational field is literally the 'space' the vehicle is IN (only the insertion-burn timing is A)
Perturbation environment & orbit sustainability (J2, drag at perigee, luni-solar, SRP)	whether the achieved orbit is sustainable under secular perturbations (1 = long-lived/bounded, 0 = transient decay or resonant growth)	perigee below the drag threshold → rapid decay; or resonant/secular growth the orbit cannot tolerate → intended orbit not maintained	secular perturbation theory (J2 nodal/apsidal rates), atmospheric drag models (NRLMSISE/Jacchia), SGP4 / numerical propagation	P (near-clean) — the perturbing field is context; only station-keeping that fights it is A
Atmospheric density profile $\rho(h)$ / layer structure along the flight path	margin between day-of-launch density actually flown and the density envelope the loads cycle was certified for	vehicle is in a density regime it was never sized for — anomalously dense low atmosphere, or an inversion/shear layer the loads analysis never bounded	reference atmospheres — US Standard Atmosphere 1976, Earth-GRAM density dispersions, day-of-launch measured profiles	pure Position state, but ρ feeds Q → into Form (loads) and Action (throttle); the density field factors clean to P, its consequences do not

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
Winds aloft, wind shear & discrete gusts (steady profile + 1-cosine gusts)	how far the measured day-of-launch wind profile sits inside the certified wind/loads envelope, and margin against shear/gust magnitude the loop can absorb	wind/shear/gust larger than certified → sustained or transient Q-α spike, TVC saturation, or structural overload near max-Q before the loop recovers	balloon/Jimsphere/lidar soundings, wind-biasing & trajectory wind-loads, discrete-gust (1-cosine) & gust-response (indicial) analysis, day-of-launch I-load updates	P input; effect is $F \times P \times A$ once the control response (A) and the resulting bending (F) are included — the wind field itself is Position
Entry corridor (undershoot /overshoot boundary for propulsive return)	normalized margin of the flight-path-angle / entry state inside the corridor bounded by peak-g/peak-heat (undershoot) and skip-out (overshoot)	outside the corridor: too steep → exceed survivable deceleration/heat rate; too shallow → skip back out and miss the reachable set	entry-corridor analysis via Allen-Eggers / Chapman ballistic-entry equations, g-load & heat-rate boundaries, bank-angle lift-modulation guidance	$F \times P$ — undershoot wall set by structural g-limit & TPS heat-rate capability (Form) against the atmospheric regime (Position); corridor geometry is Position
Downrange/crossrange footprint (reachable set) to the landing site	whether the declared landing site lies inside the reachable footprint with margin, given current state and control authority	target site outside the reachable set — no admissible trajectory connects the current state to the pad/drone-ship; the vehicle physically cannot arrive	reachability/footprint computation, bank-angle-modulation range control, predictor-corrector entry guidance, divert-envelope analysis	$P \times A$ — the reachable set shrinks with remaining propellant / L/D authority (Action); site geometry and field are Position (a named §B.0 P-A confound)
Landing-site suitability / hazard field at the LZ	existence and quality of a viable touchdown footprint — terrain slope, roughness, obstacles, pad/drone-ship position relative to approach	no safe footprint in the reachable area — the ground the vehicle can reach is a slope/obstacle/rock field it cannot survive setting down on	landing-site selection & certification, terrain-relative navigation, hazard detection & avoidance (HDA) sensing, slope/rock-abundance site models	$P \times A$ — site suitability is Position, but resolving it in real time (HDA divert to a safe spot) consumes sensing + divert authority (Action)
Touchdown state within the landing ellipse & tip-over stability cone	whether the terminal state at contact (3σ ellipse position, sink/lateral velocity, attitude, CG over gear polygon) sits inside the survivable touchdown envelope	touchdown outside the ellipse or beyond the tip-over angle / gear stroke → topple, gear collapse, or a lateral-velocity slide-out	touchdown dispersion / landing-ellipse analysis, tip-over criterion (CG ground-projection vs gear footprint on local slope), 6-DOF Monte-Carlo landing dispersions	$F \times P$ — tip-over cone is CG-height/gear-base geometry (Form) against local slope and horizontal velocity (Position); terminal control shapes the arriving state (Action)
Reachable-set membership in the gravity field (abort-to-orbit / RTLS corridor)	margin of the current state inside the reachable-set boundary (1 = deep interior, 0 = on or past the boundary)	dispersions carry the state outside the reachable set → no admissible trajectory to any target orbit remains (abort-to-orbit lost, no viable insertion)	reachability analysis, 6-DOF Monte-Carlo dispersion, abort-to-orbit / return-to-launch-site corridor design	$F/P/A$ — a genuine product: depends on remaining propellant/ ΔV (A), vehicle mass (F), and the state (P); does not factor to one bag

7.3 Action (A ↔ Energy) — thrust, control, and the release of energy

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
Chamber pressure (Pc) regulation	normalized margin of delivered Pc against setpoint ($P_c = \dot{m} c^* / A_t$) — how far Pc can wander before c^* -efficiency and wall heat-flux limits bite	chamber-pressure collapse: blow-out/flameout or throat burn-through — c^* efficiency → 0, no useful high-enthalpy gas, engine effectively off	chamber mass/energy balance and c^* efficiency, closed-loop Pc regulation via feed/throttle, steady-state engine power-balance model	$F \times A$ — Pc is fixed jointly by throat area A_t (Form geometry) and mass flow $\dot{m}$ (Action); not clean
Thrust / thrust-coefficient (C_F) delivery & engine-out	delivered thrust versus commanded, realizing $F = \dot{m} V_e + (P_e - P_a) A_e$ and its C_F	engine-out: uncommanded thrust termination on that engine — total loss of that thrust source (drives engine-out steering/abort logic)	thrust equation & thrust coefficient, nozzle gas-dynamics, multi-engine thrust management and engine-out reconfiguration logic	$A \times P$ (ambient P_a set by altitude = Position drives the pressure-thrust term) and $A \times F$ (nozzle area ratio A_e is Form)
Specific impulse (Isp) delivery	delivered Isp vs thermochemical-theoretical ($I_{sp} = c^* C_F / g_0$) — the propulsive-efficiency margin of the energy conversion	loss of propulsive efficiency: effective exhaust velocity collapses (poor combustion or gross over/under-expansion), ΔV budget unmeetable even with propellant remaining	thermochemical-equilibrium codes (NASA CEA), $c^* \cdot C_F$ decomposition, nozzle expansion / altitude-compensation analysis	$A \times P \times F$ — expansion ratio (Form) and ambient pressure/altitude (Position) both set delivered Isp; not clean
Mixture ratio (O/F) control & propellant utilization	how well instantaneous O/F stays in the design combustion band and hits the PU (residual/outage) target across the burn	O/F excursion: too lean → flameout; too oxidizer-rich → throat/wall burn-through; severe outage strands one propellant as dead mass	mixture-ratio / propellant-utilization control, adiabatic-flame-temperature thermochemistry, closed-loop MR control, residual/outage Monte-Carlo	$F \times A$ — injector pattern and feed-line impedances plus trapped-residual tank geometry (Form) set achievable O/F and unusable propellant
Deep-throttle authority (landing-burn / hoverslam regime)	commandable thrust-range margin down to hoverslam thrust while staying above the low-Pc combustion-stability boundary	throttle stuck or injector unstable at low ΔP (stiffness collapse) → cannot reach required thrust → hoverslam miss / hard impact	throttle-valve control, low-Pc injector stability boundary ($\Delta P/P_c$ collapse), combustion-stability map vs power level	$A \times P$ — the landing-burn corridor/timing are Position-bound and Action margin is thinnest at near-depletion; min-thrust vs near-empty weight is $F \times P \times A$
Propellant / ΔV budget & landing-fuel reserve	remaining usable propellant against required ΔV plus reserve (rocket equation), including predicted residual/outage	propellant depletion before end of burn — thrust ends with a ΔV shortfall or no landing margin, even with a healthy engine	Tsiolkovsky rocket equation, vehicle mass/ ΔV budget, propellant-residual (outage) prediction and gauging, powered-descent reserve sizing	$A \times P \times F$ — ΔV demand is trajectory-set (P) and usable-vs-residual propellant is set by tank/ullage geometry (F); the landing curve is $A \times P$

Detail	Score measures (0-1)	Collapse (pillar→0)	Governing operational theory	Coupling
TVC control authority (gimbal deflection/rate vs required control moment)	fraction of the commanded restoring moment the gimbal can deliver — margin between demanded gimbal angle/rate and the actuator hardstop/rate limit	TVC saturation: gimbal pinned at deflection or rate limit, no incremental control moment available, attitude diverges and the vehicle tumbles	control-authority / disturbance-moment budget, actuator rate & deflection limits vs classical rigid-body autopilot demand, 6-DOF Monte-Carlo	F×P demand met by A — required authority set by CP–CG static margin (F×P) and dynamic pressure Q (P); only the actuator supply is Action
Guidance law / powered-trajectory tracking (ascent PEG; convex landing guidance)	convergence & feasibility margin of the closed-loop guidance solution — whether the commanded thrust profile still reaches the required end-state	guidance fails to converge / commands an infeasible burn: targets missed, ΔV squandered, vehicle leaves the reachable set	Powered Explicit Guidance / optimal control; for landing, convex trajectory optimization / successive convexification (G-FOLD-class)	A×P — guidance is the operator mapping thrust/energy (Action) onto trajectory and reachable set (Position); does not factor to Action alone
Navigation / state estimation (IMU + GNSS/star-tracker nav filter)	estimated-state accuracy vs the navigation error budget — filter covariance/innovation consistency and observability vs required tolerance	nav divergence (IMU failure / filter divergence): guidance and control act on a wrong state, commanding the vehicle out of the corridor while believing it nominal	Kalman/EKF filtering, INS error-propagation & covariance, GNSS-INS integration, observability Gramian, innovation chi-square, sensor FMEA/redundancy	A→P — the estimate is an Action-side sensing loop, but what it estimates is Position; a nav error is indistinguishable downstream from a true Position excursion
Reaction-control-system (RCS) attitude control — coast / exo-atmospheric	RCS torque authority and remaining thruster propellant vs the attitude-correction and deadband-limit-cycle demand where aero and TVC are unavailable	RCS propellant depletion or thruster-out: no attitude authority in vacuum, the vehicle drifts off commanded attitude with no recovery	phase-plane / bang-bang (deadband) control, limit-cycle & duty-cycle analysis, RCS propellant budgeting, thruster FMEA	clean — genuinely factors to Action (control torque + capacity), aside from shared propellant/capacity accounting
FDIR core — fault detection, isolation & recovery / redundancy management	fault coverage × correct-isolation probability minus false-alarm penalty (1 = faults detected, isolated, reconfigured in time; 0 = undetected fault OR spurious trip of healthy hardware)	symmetric failures: a real fault goes undetected and propagates to loss of vehicle, OR FDIR falsely trips good hardware and removes control authority it should have kept	FMEA/FTA, Markov reliability for N-modular-redundant strings, voting/mid-value-select, Byzantine-fault-tolerant agreement, time-to-criticality budgeting	clean-A as logic — the meta-Action pillar that arbitrates faults arising in F and P; its own hardware hosting couples weakly to Form
Abort / flight-termination trigger logic & phase sequencing	decision-margin of the trigger set: correct fire within the abort window vs false/late trip, per phase (1 = triggers armed and correct at every phase boundary, 0 = missed window or spurious termination)	a missed abort carries a survivable state into loss, OR a spurious flight-termination destroys a healthy vehicle	abort-trigger / red-line design, autonomous flight-safety rule sets, RCC 319/321 range-safety criteria, phase-transition (state-machine) verification, hazard analysis	A×P — trigger thresholds are defined against the trajectory corridor and instantaneous-impact-point (Position); the decision machinery itself is Action

7.4 The cross-pillar rows — where separability is genuinely at risk (the TRA-P0 confound set)

Phenomenon	Spans	Why it does not factor to one bag
POGO — structure ↔ feedline ↔ thrust longitudinal loop	F×P×A	Axial structural mode (F) closes with feed-system/combustion dynamics (A) at an acceleration/structural-mode state (P) into one inseparable loop; the failure IS the coupling, so no single pillar is near zero and U reads high right up to a divergent limit cycle. Certified by POGO stability analysis (Rubin transfer-matrix + accumulator sizing), not a scalar.
Propellant slosh coupled into the attitude-control loop	F×P×A	Tank/baffle geometry (F) sets slosh modes whose natural frequency scales with axial acceleration (P) and whose danger is coupling into the GNC loop (A); worst near depletion. Governed by mechanical-analog slosh models + baffle damping + gain/phase margin at the slosh frequency.
Aeroservoelasticity / control-structure interaction (bending mode × Q × loop)	F×P×A	Mode shapes/frequencies (F, migrating as propellant drains) are excited with growing dynamic pressure (P) and closed by the autopilot (A); the loop can pump energy into a bending mode to divergence. Handled by notch-filter / mu-analysis over FEA modes, GVT, and flutter boundaries — objects U has no representation for.
Combustion- acoustic high- frequency instability (screech/HF)	F×A (double -loads Form)	Chamber/injector geometry (F) sets the acoustic mode shapes; unsteady combustion (A) drives them; filed to Action but every fix (baffles, resonators, injector face) is geometric. Rayleigh criterion + Crocco $n\text{-}\tau$ + acoustic-mode analysis predict onset in milliseconds; a geometric mean cannot represent a mode phase.
Static margin (CP- CG separation)	F×P → tipping to P×A	A CP-CG geometry (F) that only has meaning in an aerodynamic regime (P); when it goes negative all restoring authority shifts to TVC (A). The appendix's explicit named §1.2/B.0 confound — U cannot produce the sign, the caliber, or the CG shift that fixes it.
Closed-loop gain / phase (pitch-yaw) stability margin	F×P×A	The plant is set by bending-mode shapes (F) and aero derivatives at dynamic pressure Q (P), closed by the autopilot (A); the appendix files it as an emergent F×P×A product. U is a lossy projection of exactly the multi-dimensional, frequency-resolved margin surface Nyquist/mu resolve.
Q-alpha / Q-beta aerodynamic bending load (max- Q)	F×P×A	GNC attitude (A) × dynamic pressure (P) × airframe bending capability (F) form an emergent product; the placard is Form, the environment is Position, the response is Action. Load-relief steering and Q-alpha dispersions govern it, not a separable index.
Stage-separation / phase-boundary event	F×P×A	Joint/interstage geometry (F) × relative attitude-rate and residual aero at the sep point (P) × sep impulse/retro/ullage thrust and timing (A) are genuinely non-separable, with an instantaneous mass-property discontinuity. Multibody 6-DOF sep dynamics + pyroshock + Monte-Carlo tip-off own it.
Landing-burn coupled hard curve	A×P (with F at touchd own)	Near-depleted thin throttle/ ΔV authority (A) meets the violent narrow entry/landing corridor (P) and they collapse together; at contact the near-empty dry mass (F) in the local g-field sets the tip-over margin. §2's Action-margin-bound-and-Position-corridor-bound curve — the two floors move as one.
Nozzle flow separation & side- loads	F×P	Separation onset is set by ambient back-pressure/altitude (P) against a fixed bell contour (F); over-expansion at sea level drives destructive lateral side-loads. A geometry-in-a-context confound, not clean Form — governed by Summerfield / Kalt-Bendall separation criteria.

These are exactly the phenomena §B.0 conditions on and the perturbation-leakage metric must register. **The matrix is honest about its own leaks:** the rows marked with an F/P/A coupling are where "three separable bags" is a claim under test, not a given.

8. Relation to operational rocket-engineering theory — U sits ABOVE the specialists, it does not replace them

$U = \sqrt[3]{(F \cdot P \cdot A)}$ **does not compete** with any established stability theory below. For the thing each one does, **the specialist theory is strictly better and TRA defers to it** — U owns no laboratory, no proof, no margin of its own (§0, §4). U's only *proposed* contribution is to sit **above** them as a single, cross-cutting, non-compensatory survival index that makes "*which pillar is the floor*" legible across all of them at once — a triage/communication layer, gated by §B, never a control law. Where the relation reads **subsumes_U**, U is redundant there and says so.

Operational theory / tool	What it is & does	Relation to U	Honest verdict (where U helps · where the specialist is strictly better)
Nonlinear / Lyapunov stability theory (Lyapunov functions, region of attraction, ISS)	Rigorous dynamical-systems certification: a $V(x)>0$, $\dot{V}(x)<0$ supplies a constructive stability proof over a neighborhood; sublevel sets give certified region-of-attraction (basin) estimates via SOS/LMI; ISS bounds the state by class-KL decay plus a class-K disturbance gain. Anchors CLF/CBF-QP safety filters and barrier certificates. Quantitative, per-system, dynamics-dependent — produces certificates and guaranteed sets.	competes	U sits strictly downstream. Lyapunov delivers what U structurally cannot: a certificate ($\dot{V}<0$), a certified basin of recoverable states, ISS disturbance guarantees, and hard barrier constraints — and it captures the POGO/slosh/ASE coupling inside one V that U's separability premise fails on. U's only defensible niche is ABOVE the stack: a non-compensatory cross-domain triage index answering 'which pillar is the floor now' across incommensurable margins — a question a per-mode certificate deliberately does not answer. Forbidden from replacing certification, entering any loop, or standing in for a region of attraction.
Linear & robust control margins: Nyquist/Bode/Nichols, gain/phase/disk margin, LQR/LQG, H-infinity, structured-singular-value (μ) analysis	A mature per-loop theory of dynamical stability/robustness for linearized systems. Nyquist decides closed-loop stability by encirclement counting; gain/phase margins read tolerable gain growth and phase lag; H_∞/μ model plant uncertainty explicitly and synthesize or certify worst-case robust stability. Delivers control laws that run in the loop, predict instability frequency and onset, and are the certification-grade tools for flutter, POGO, slosh, ASE, and rigid-body attitude.	subsumes U	Robust control decisively subsumes U on everything dynamical and is largely orthogonal elsewhere. Its margins are frequency-resolved, per-mode, in-the-loop, theorem-backed; U is a lossy scalar projection of exactly the multi-dimensional margin surface the appendix files as an emergent $F \times P \times A$ product. U can never be a stability margin, never in a loop, never a Nyquist/ μ certificate. Its only out-of-loop value: flagging when the binding survival constraint is NOT a control-loop problem at all (structure or corridor) — a coarse pointer, useless as a margin, dangerous if mistaken for one, and unproven until TRA-P1/P2 pass.
Aerodynamic / flight stability: static margin (CP-CG), damping derivatives, dynamic stability (eigenvalue/modal) analysis	Physics-based theory of whether a vehicle holds and returns to a commanded attitude: static margin (CP-aft-of-CG restoring stiffness), stability & damping derivatives (C_{m_alpha} , C_{n_beta} , C_{m_q} , C_{n_r} , Cl_p), and dynamic stability from linearized 6-DOF eigenvalues (short-period, phugoid, Dutch-roll, aeroelastic modes) yielding damped frequencies, damping ratios, and time-to-double/half. Predictive, units-bearing, certifiable.	subsumes U	On vehicle stability the established theory subsumes everything U could say, and U concedes it — the appendix files static margin, pitch/yaw margin, and gain/phase margin as an emergent $F \times P \times A$ product, i.e. a coupling that smears across all three bags, so U cannot decompose or derive it. Aero stability gives a number, a sign, a frequency, a damping ratio, a certifiable threshold (negative static margin, min damping); U's cube-root is compensatory in the interior and carries no units or eigenstructure. U's only add is a coarse cross-domain 'which pillar is the floor' label, and only if TRA-P1/P2 pass.

Operational theory / tool	What it is & does	Relation to U	Honest verdict (where U helps · where the specialist is strictly better)
Structural-coupling instabilities: POGO, propellant slosh, aeroelasticity / ASE / flutter	A quantitative dynamical theory of closed-loop feedback BETWEEN subsystems — energy-exchanging loops solved as governing PDEs/ODEs in the frequency/eigenvalue domain (Nyquist, p-k / V-g flutter boundaries, gain-phase margins), validated by GVT, wind-tunnel flutter tests, and flight telemetry. Outputs the instability onset condition (flutter speed, POGO gain margin, slosh phase margin), the frequency, the mode shape, and the concrete detuning fix.	orthogonal	The sharpest counterexample to U's premise, honestly conceded: these failures ARE the F×P×A coupling, not any one pillar, so U — built on separable bags — is structurally blind. Flutter/POGO kill the vehicle when F, P, A are each individually healthy, so U reads high right to divergence and its zero-limit never triggers. Orthogonal: the instability theory emits no survival scalar, U emits no stability boundary. Worse, U's argmin 'floor pillar' routing is actively misleading here — there IS no floor pillar; pointing at one bag hides the interaction that is the threat. Against this class U should be reported as failing TRA-P1, not spun.
Operational ascent/entry/landing GN&C stack: trajectory optimization (Pontryagin/colllocation/pseudospectral, convex G-FOLD), 6-DOF Monte Carlo dispersion, LinCov, closed-loop optimal guidance	A quantitative, physics-closed, units-bearing, certification-grade toolchain: it generates the reference trajectory, propagates the true coupled 6-DOF equations with time-varying mass properties (propellant drain moving CG and bending modes), returns a calibrated P(loss) and dispersion ellipsoid, enforces hard constraints (max-Q, static-margin floor, landing-fuel reserve) as real thresholds, models POGO/slosh/ASE by coupled dynamics, and — in the guidance case — flies the vehicle.	subsumes U	Subsumes U on everything U claims about survival: MC returns a calibrated P(loss) where U offers a dimensionless index with a placeholder ϕ^{-1} threshold; LinCov returns actual distance-to-corridor-edge covariance; guidance actually acts; MC's OR-over-disciplines already IS zero-limit non-compensation, calibrated, coupled, in native units. U's factorization is weakest exactly where rocketry is hardest (the F×P×A confounds). The thin non-subsumed residue: no single tool natively emits ONE cross-discipline scalar pointing at 'which pillar is the floor' — a dashboard convenience largely recoverable from MC/Sobol sensitivity, with zero predictive content until TRA-P1 passes.
Reliability & safety engineering (FMEA/FMECA, FTA, PRA/PSA, redundancy & fault management)	A standardized discipline (MIL-STD-1629A, IEC 60812/61025, NASA NPR 8705, NUREG-1150): FMECA enumerates failure modes bottom-up by severity×occurrence×detection; FTA works top-down through Boolean gates to minimal cut sets; PRA composes these with event trees and common-cause models into a calibrated LOV/LOC probability with uncertainty. Causal, auditable, certifiable, consuming real failure-rate data; it gates flight readiness.	complements	U sits above and outside as a coarse triage index and is subsumed on everything certification-relevant. Its one marginal value — the argmin 'which pillar is the floor' pointer — is honestly a restatement of the FTA idea that an AND-gate through the min-availability branch dominates the cut set. Reliability engineering is strictly better: calibrated probabilities vs U's uncalibrated scalar with a physically-meaningless 0.618 placeholder; cut sets capture combinatorial/common-cause failures where U's product collapses everything into three compensatory pillars; FMECA/FTA handle the cross-pillar confounds natively without any separability premise; and they trace to a specific component and fix while U only proposes an address a specialist must confirm.

Operational theory / tool	What it is & does	Relation to U	Honest verdict (where U helps · where the specialist is strictly better)
Systems-engineering margin management & MBSE — TPMs, margin/allocation stack-up, model-based verification (NASA SP-6105/NPR 7123, INCOSE, ANSI/EIA-632, ISO/IEC 15288, ECSS)	The operational discipline that keeps a program feasible: it defines Technical Performance Measures (mass, ΔV , power, thermal, structural FoS, propellant reserve, gain/phase margin), assigns each a required value plus planned margin, stacks per-discipline contingency/MGA, tracks burn-down against a waterline with a named owner, and closes each by verification traceable through a requirements tree. MBSE keeps allocation, margin state, and ICDs live and queryable. Deliverables are per-parameter, hard-constraint, auditable, certification-grade.	complements	U complements as a coarse cross-domain readout but is subsumed quantitatively. Value: one glance at $\pi^* = \text{argmin}(F, P, A)$ surfaces which family is driving loss-of-vehicle risk where a flat margin waterfall does not; zero-limit non-compensation correctly encodes that surplus in two families cannot buy back a collapsed third. But margins are hard constraints with units and thresholds — a Q-limit, positive static margin, min landing-fuel reserve — and U is compensatory in the interior, so it is the wrong object anywhere safety is a floor not a trade; the real killers are $F \times P \times A$ couplings that violate its separability premise; and collapsing many owned, dimensioned margins into one scalar destroys the accountability that makes TPM tracking work. U is a lightweight overlay, never the system of record.

Net (honest): $U = \sqrt[3]{(F \cdot P \cdot A)}$ is a coarse, dimensionless, cross-cutting survival/attention index that SITS ABOVE the specialist aerospace toolchain and never replaces any part of it. Its only defensible content is two things: (1) zero-limit non-compensation — a truly dead pillar (structure gone, out-of-corridor, thrust gone) makes survival zero and no surplus elsewhere buys it back, encoding cross-discipline non-substitutability that no single specialist theory states in a common currency; and (2) argmin routing π^* that flags which of Form/Position/Action is the current floor across incommensurable failure classes, a triage/communication pointer for design reviews. Everything quantitative is owned by the incumbents and they are strictly better where a number is needed: Lyapunov/ISS give certificates and regions of attraction; Nyquist/ μ give frequency-resolved in-loop margins; aero stability gives static margin, damping ratios, and time-to-double; 6-DOF Monte Carlo gives a calibrated $P(\text{loss})$; FMEA/FTA/PRA give cut sets and calibrated failure probabilities; TPM/MBSE gives owned, dimensioned hard constraints. U is compensatory in its interior ($F=0.5, P=1, A=1 \rightarrow U=0.79$), so it can never represent a hard threshold — a max-Q limit, a static-margin floor, a gain/phase margin, a landing-fuel reserve — and it is explicitly barred from any control loop, from arbitrating a constraint, and from standing in for a certified margin. Its separability premise is weakest exactly where rocketry is deadliest: POGO, slosh, aeroservoelasticity, combustion-acoustic instability, static margin, and closed-loop gain/phase margin are irreducible $F \times P \times A$ couplings where the failure IS the coupling — no pillar is near zero, so U reads high right up to divergence and its zero-limit never fires. Those phenomena are quarantined into cross_pillar_rows precisely so U is not trusted to score them. The value claim remains unproven: TRA-P0 (does separability even hold on this vehicle) and TRA-P1/P2 (U must beat a fitted margin-ensemble, a compensatory mean, and a free-exponent power-law fit, stratified by failure class) are unrun, and against the structural-coupling class U is expected to fail. Bottom line: use U as a space-infrastructure triage OVERLAY / attention router — with any threshold near 0.75 being an UNVALIDATED SSS stakes-scaled default, fit-or-discard under TRA-P1 and NEVER a safety-of-flight gate — built on top of margins the specialists already compute rigorously — useful as a pointer, useless as a margin, and dangerous if ever mistaken for one.

Governing limitation (REVIEW-v2 M6). The modes that *cleanly factor* to one pillar (rotordynamics, welds, RCS, fatigue, orbital perturbations) are exactly the independently-instrumented modes the incumbents already own; the historically *dominant* loss-of-vehicle mechanisms are the **coupled** ones (§7.4). So 'where TRA separates cleanly' and 'where TRA adds value' are physically **near-disjoint** — TRA's honest niche is the cross-cutting overlay, not the coupled failures.

9. SSS integration — TRA is a client of the System Stability Score engine

TRA does not re-implement scoring; it is a **conformant client of SSS** ([APPENDIX_SSS](#)), the universal measurement engine. Concretely:

- Core scoring: TRA consumes the SSS engine's $U = \sqrt[3]{(F \cdot P \cdot A)}$ verbatim (SSS.2), with the argmin router $\pi^* = \text{argmin}(F, P, A)$ naming the current 'floor' pillar and δ (the F/P/A imbalance/dispersion) plus SI (the imbalance-penalized readout; legacy label 'stability index') supplied as the secondary readouts TRA routes on — U for the one-glance verdict, δ/SI for which pillar is eroding and how fast.
- Threshold gate: TRA does NOT use the 0.618 cross-domain default. Per SSS.2's explicit stakes-scaling ('space infrastructure ≥ 0.75 , nuclear ≥ 0.90 '), the SSS stakes-scaled DESIGN-REVIEW DEFAULT for space infrastructure is $U \geq 0.75$ (vs 0.618 for ordinary Earth) — but TRA claims NO numeric acceptance gate: 0.75 is UNVALIDATED, identical in fit-or-discard status to 0.618, may be spoken only AFTER TRA-P1, and 0.618 is NOT declared 'fail' for flight.
- F/P/A/ Σ jury: the four TAA agent roles (Form, Position, Action, Generalizing- Σ) are the SSS jury structure (SSS.3 Stage 2, TAA mapping). The F/P/A agents score the matrix rows against specialist ground truth; the Σ agent enforces the non-compensatory geometric logic and audits separability, quarantining the cross-pillar confounds so no coupled instability is scored inside a single bag.
- Measurement engine: the SSS Constructor (Stage 1) generates Form.md / Position.md / Action.md — these files ARE the consolidated matrix rows; the up-to-50-model jury (Stage 2) scores each row 0–100, IQR-filters outliers, takes per-principle weighted means → pillar averages → geometric mean, exactly as SSS runs for a heart or a nation, only the principles change.
- Continuous-loop wiring (SSS.8.5): TRA calls SSS at three LGP sync points — LGP-1 Scanning for baseline U_0, δ_0 ; LGP-10 Pulse Monitoring for SI-dot and δ -volatility V_δ via EWMA/CUSUM (a **design-review / simulation attention flag** to the TAA agents if SI is declining — **never a flight-abort signal**); LGP-12 Final Audit for final U_f and $\Delta\text{SI} = \text{SI}_f - \text{SI}_0$ that inform the **analysis** review priorities (**never a live flight gate**).
- Execution order & guard: TRA follows the canonical GSI-RTD → TAA → LGP-12 → SSS pipeline, and routes **post-hoc analytic** results through the SSS-Guard ensemble (3 independent SSS instances; a ≥ 2 -agree rule as a **variance-reduction convenience for the analysis, not a flight verdict**) — operating in **simulation / design-review only**, with **NO authority over flight-readiness, abort, continue, certification, or constraint arbitration**. U audits but never enters the control loop, never arbitrates a hard constraint, and never substitutes for a certified margin; **all flight decisions are owned by the specialist tools and human sign-off**.

This resolves REVIEW minor-3 (the φ^{-1} placeholder). SSS itself declares the threshold domain- and stakes-calibrated, and names **space infrastructure ≥ 0.75** (vs the ordinary-Earth default 0.618). TRA therefore inherits an **unvalidated, stakes-scaled design-review default (≈ 0.75) from SSS** — not a bare φ^{-1} , and not a claimed survival gate — and even that number remains subject to fit-or-discard under TRA-P1. Execution order is inherited unchanged: **GSI-RTD $\rightarrow$ TAA (F/P/A/ Σ jury) $\rightarrow$ LGP-12 (cycle) $\rightarrow$ SSS (audit).**

10. Multi-model construction & audit of the matrix (the "multi-intellect" layer)

The matrix is built and audited by the four-role TAA jury mapped onto the pillars — a Form agent, a Position agent, an Action agent, and a Generalizing agent (Σ) — instantiated as a multi-model panel (per SSS Stage 2, up to ~ 50 OpenRouter models scoring in parallel, IQR-filtered, per-principle weighted mean $\rightarrow$ pillar average $\rightarrow$ geometric mean). Construction runs in the canonical execution order GSI-RTD $\rightarrow$ TAA $\rightarrow$ LGP-12 $\rightarrow$ SSS: GSI-RTD supplies the recursive search space and the scheduler that decides WHEN each subsystem is measured; TAA's Constructor stage emits the three cross-sectional files (Form.md / Position.md / Action.md) whose falsifiable principles ARE the matrix rows above — each row carries a `scores_what` in $[0,1]$, a `collapse_mode` (the P/F/A $\rightarrow$ 0 zero-limit event), and a `governing_theory` that names the real specialist tool that owns the number; LGP-12 (Final Audit) is where SSS scores the assembled rows and computes U , δ , and ΔSI . The division of labor is deliberate: the F/P/A agents score WITHIN their pillar against the specialist ground truth, while the Σ agent's sole job is to police separability — it flags every row whose coupling field is not clean-single-pillar and routes the genuine $F \times P \times A$ confounds (POGO, slosh, ASE, static margin, gain/phase margin, Q-alpha, staging, the landing-burn curve) OUT of the per-pillar bags and INTO `cross_pillar_rows`, because scoring them inside one pillar would let U read high while the coupled loop diverges. The multi-model panel **reduces single-instance variance** (no single model's domain bias dominates — but shared training data and correlated bias mean this is *variance reduction, not proven independence or impartiality*, and must be **measured, not assumed**), and the SSS-Guard ensemble (3 independent SSS instances, ≥ 2 -agree tolerance) operates in **simulation / design-review only** — it mirrors the hard firewall that U is a scorecard, never a control input, and it holds **no flight-readiness, abort, or certification authority**. The audit is adversarial by design: TRA-P0 tests whether the separability premise even holds on this vehicle, and TRA-P1/P2 pre-register that U must beat a fitted margin-ensemble, a compensatory mean, AND a free-exponent power-law fit (baseline iv) of the same specialist margins, stratified by failure class — a bar the Σ agent expects U to FAIL specifically on the structural-coupling class, which it must report rather than spin.

Discipline, not magic (per RH): a multi-model jury constructs and cross-checks the matrix faster and with less single-author bias than one engineer — but it produces **claims with provenance and a B-level stamp, not ground truth**. Every matrix cell is admissible only with its *Governing operational theory* citation; a cell without a specialist tool behind it is marked `axis_unseparated` and is inadmissible to any downstream U . The jury sets the queue; it never signs the physics.

11. GSM — modelling the F×P×A coupling instead of pretending it away (uncertainty-aware $\mathbf{U}$; borrowed from TRC §2.9 via the TE β -channel)

The §7.4 cross-pillar rows (POGO, slosh, aeroservoelasticity, combustion-acoustic, static / gain-phase margin) are exactly where "three separable bags" is false and **the failure IS the coupling** — no pillar is near zero, so a point $\mathbf{U}$ reads high right up to divergence. TRA does not pretend these away; it borrows the **Gaussian Stability Matrix (GSM)** from [APPENDIX_TRC §2.9](#):

- Stack the pillar/phase log-odds into $\mathbf{z} \in \mathbb{R}^n$, μ mean, $\Sigma = \mathbf{D} \cdot \mathbf{R} \cdot \mathbf{D}$. The **off-diagonal correlation blocks $\mathbf{R}$** are the F/P/A dependence — the same ρ_{FPA} TRC-P0 measures — **estimated, never assumed**, ingested from the **TRA-P0 perturbation-leakage battery** (§B.0): the measured leakage Λ informs the off-diagonal via a **stated, pre-registered monotone mapping $\rho = m(\Lambda)$** — a **modelling choice, NOT an identity** (Λ is a non-negative distance-dependence; ρ a signed Gaussian correlation, so the sign and calibration of m are declared, never assumed). $\mathbf{R}$ is shrunk to a PSD estimate (Ledoit–Wolf / OAS); and **when dependence is under-determined, the decision-use rule is bounds / fail-closed, not a point estimate** — shrink-to-independence is used only for visualization, since it can *under-state* common-cause risk.
- $\mathbf{U}$ is then **not a point but a credible interval reflecting the modelled dependence** — positive coupling generally widens it and drops the lower tail. POGO / slosh / flutter therefore no longer read as a false-high point $\mathbf{U}$; they read as a **wide, low-confidence $\mathbf{U}$** . This is **epistemic uncertainty under a stated covariance assumption — a risk-visualization, NOT an instability detection or a stability certificate**: a wide interval says the *model is unsure*, not that divergence is proven.
- Every off-diagonal carries `G{source; test; date}` provenance (§10 discipline); a Σ with assumed off-diagonals is `axis_unseparated` and renders no interval, only the $\Sigma \rightarrow 0$ point fallback.
- Risk = a model-based failure-probability estimate** under the stated GSM assumptions (an unvalidated model output, **not** a validated posterior) against the (unvalidated, stakes-scaled) threshold; **GSM-Priority** = $E[\Delta P_{\text{fail}} \mid \text{intervene on cell}] / \sqrt[3]{(C_{\text{time}} \cdot C_{\text{space}} \cdot C_{\text{energy}})}$ — a *min / posterior* object, not a mean.

11.1 Non-compensatory within-pillar aggregation (resolves REVIEW-v2 M1)

The §7 grid holds ~12 rows per pillar. **A weighted mean is wrong for loss-of-vehicle (LOV) rows**: a single common-bulkhead rupture, COPV burst (AMOS-6), weld-defect tank rupture, single-engine RUD, or throat burn-through zeros **one** row while the rest stay healthy — a mean keeps the pillar ≈ 0.9 and $\mathbf{U}$ stays high, so the very catastrophes §2 cites as the zero-limit are **not reproduced by the appendix's own aggregator. v1.3 rule**: LOV-class rows aggregate **non-compensatorily** — the pillar takes a **hard per-row floor / min over the LOV set** (one dead LOV row drives its pillar $\rightarrow 0$), while non-LOV rows may mean-aggregate. This makes "the zero-limit is always visible" **true of the aggregator**, not merely of an idealized fully-collapsed pillar. The LOV-row set is the pre-registered FMEA / FTA catastrophic set, not chosen post hoc.

11.2 GSM in practice — a runnable reference, real numbers, and the "GSM-lite" recipe

GSM is not just prose: `tra_gsm_reference.py` (shipped with this appendix) implements it in ~30 lines of NumPy and runs in a second. Two results make the mechanism concrete (illustrative pillar inputs; the $\mathbf{U}$ -distribution math is exact):

Coupling widens the interval and drops the tail. For an imbalanced state ($F=0.90$, $P=0.55$, $A=0.90$), raising the cross-pillar correlation from $\rho=0.10$ to $\rho=0.65$ widens the 90% credible interval of U from $[0.731-0.793]$ to $[0.725-0.799]$ and lowers $P(U \geq 0.75)$ from **75% $\rightarrow$ 72%** — the point estimate is unchanged (0.764) but the *risk* rose. That is the physics of coupled failure the scalar hides.

The distributional gate flags the right phase. Scoring a Starship-class phase profile against the rule "*pass iff $P(U \geq 0.75) \geq 0.90$* " — this answers the "how do you threshold a distribution" question: you gate the **probability of exceedance**, not a scalar:

Phase	point- U	90% CI	$P(U \geq 0.75)$	π^*	gate
Ascent (nominal)	0.900	[0.884–0.913]	100%	A	PASS
Max-Q ($F \leftrightarrow P$ aeroelastic)	0.827	[0.796–0.853]	100%	P	PASS
Staging (all coupled)	0.823	[0.792–0.849]	100%	A	PASS
Landing burn ($A \leftrightarrow P$)	0.768	[0.731–0.801]	79%	A	FLAG

The landing burn is the **only** phase the distributional gate flags — precisely the regime where the physics is hardest (near-depleted propellant + a narrow corridor). The scalar $U=0.768$ looks "fine"; the *distribution* correctly says otherwise.

GSM-lite — the practical tiering:

- 1. **Daily work:** point U + δ (cheap, scalar).
- 2. **High-risk phases (max-Q, staging, landing) or high δ** : run GSM — the interval + $P(U \geq thr)$.
- 3. **Use it as a risk visualization (plot the U distribution), not a hard decision number** — consistent with §4/§5 (U is never a control law or an acceptance gate).

12. Center–periphery by flight phase — the binding pillar migrates (a phase-indexed π^* profile)

$\pi^* = \operatorname{argmin}(F, P, A)$ is **not static** across a flight; the **center** (the binding pillar — the corpus's center–periphery stability distribution) migrates by phase, and TRA reports it as a **profile**, not a scalar. This is the design-review use of TRA — *which pillar owns the margin in which regime* — and it is itself testable (does the pre-registered center match where telemetry shows margin actually ran out?):

Phase	Center (binding pillar)	Periphery	Why
Liftoff / early ascent	Action (thrust / control authority)	Position	thrust must beat gravity with margin
Max-Q	Form (airframe under combined Q -load) $\times$ Position (Q - α corridor)	Action	structure is the strict center; aero is the driver
Staging	all three at a boundary (δ spikes)	—	separation stresses Form, attitude, and relight at once
Coast / orbit	Position (regime / injection state)	Form, Action	the vehicle is mostly a <i>place</i> ; little is done
Entry	Form (TPS / loads) $\times$ Position (corridor / heating)	Action	aerothermal-structural survival
Landing burn	Action (throttle / landing-fuel margin) $\times$ Position (reachable ellipse)	Form (loads past peak)	near-depleted propellant + narrow corridor

13. Coherence — the Kuramoto layer for multi-engine and combustion synchrony (the NDT 5th-currency analogue, made runnable)

A 33-engine cluster (Starship-class), combustion-acoustic modes, and structural modes are **coupled oscillators**: they can lock in phase — desirable thrust synchrony, or destructive POGO / screech. The **Kuramoto order parameter** $r \in [0,1]$ is the *measurable* coherence quantity (the classical analogue TRA marks provisional against NDT's quantum-reserved 5th currency). `tra_kuramoto_reference.py` (shipped) computes it.

Coherence onset. For $N=33$ oscillators with unit-std natural-frequency spread the critical coupling is $K_c = 2\sqrt{2/\pi} \approx 1.60$; below it the cluster is incoherent; above it, it locks:

K	r	state
0.5	0.30	incoherent
1.6 (K_c)	0.68	partial
3.0	0.89	LOCKED
4.0	0.95	LOCKED

A **detuned "rogue" engine** (1 of 33 far off-frequency) stays out of the lock at $K=1.6$ ($r=0.53$) but is pulled in by $K=3.0$ ($r=0.92$) — a concrete read on whether a mis-tuned engine desynchronizes the cluster or is entrained.

The §11↔§13 bridge — coherence drives the coupling. The elegant unification: the Kuramoto r feeds the GSM off-diagonal ρ_{FPA} ($\rho \approx \rho_{max} \cdot r$). As engines / modes synchronize (r rises), cross-pillar coupling rises, the GSM U -interval widens and its lower tail drops (§11.2). **Rising coherence is rising risk, computed end-to-end** — a phase model → a correlation → a survival-index distribution. This is the one place the corpus's "coherence" currency becomes a *running number*, not a promise.

Reference implementations: `tra_gsm_reference.py`, `tra_kuramoto_reference.py` (MIT; ship with this appendix).

Falsifiability matrix (one-glance)

Load-bearing claim	Falsifier	Kill threshold	If killed, TRA becomes
A flight state factors into F/P/A (net of coupling)	TRA-P0	$\Lambda \geq 0.15$, or $\leq$ random partition, or in the 0.6–0.8 band	its separable-scoring claim struck (a <i>coupled</i> coordinate system may remain, §B.0)
Rocketry is the <i>most</i> separable domain	B.0-rank	separation score does not exceed TRB/TRC/TSE by margin m	a bare B0 conjecture (no indicative use)
U beats a fitted-ensemble and a compensatory aggregate	TRA-P1	CI ≥ 0 vs. baseline (ii) or (iii)	a relabeling of known margins
The cube-root specifically carries the gain (equal- $\frac{1}{3}$ exponents)	TRA-P1, baseline (iv)	CI ≥ 0 vs. baseline (iv)	'multiplicative beats additive'; equal- $\frac{1}{3}$ exponents not special
δ / SI add predictive value over U	TRA-P1b	CI ≥ 0	communication aids only, zero evidentiary role

Load-bearing claim	Falsifier	Kill threshold	If killed, TRA becomes
Weakest-pillar dispatch beats uniform at equal budget	TRA-P2	no win at equal budget	an index, not an orchestrator
The F/P/A language reliably codes real accidents & the §12 phase profile fits them (descriptive rung below P1)	TRA-S0	low inter-rater κ , or the phase→ π^* distribution mismatches §12	a description raters cannot reproduce; the language claim is struck
Dual-use is <i>contained</i> (NOT "proven safe")	— (ungated)	—	firewall §5 holds regardless; containment is a policy boundary, not a safety proof

References & provenance

- Parent record: U-Theory, DOI **10.17605/OSF.IO/74XGR**; code repository `UniversalModel/core`.
Review of record: `REVIEW_TRA_TRIADIC_ROCKETRY_ASTRONAUTICS.md` — the **internal multi-model editorial hardening record**, v1.0→v1.5 (adversarial LLM panels, 7 reviewer lenses per round — not human peer review; the v1.1 revision addressed all 7 majors + 12 minors).
- Inherited primitives: **SSS** (score), **GSI-RTD** (runtime + weakest-pillar scheduler + β), **TE** (umbrella), sibling instances **TRB / TRC / TSE**, dual-use venue **WAR**.
- Standard aerospace ground truth (aerodynamics/CFD, structural mechanics/FEA, orbital mechanics, combustion, GNC) is *consumed, not restated*.

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